Idea Presentation

GROUP - 9

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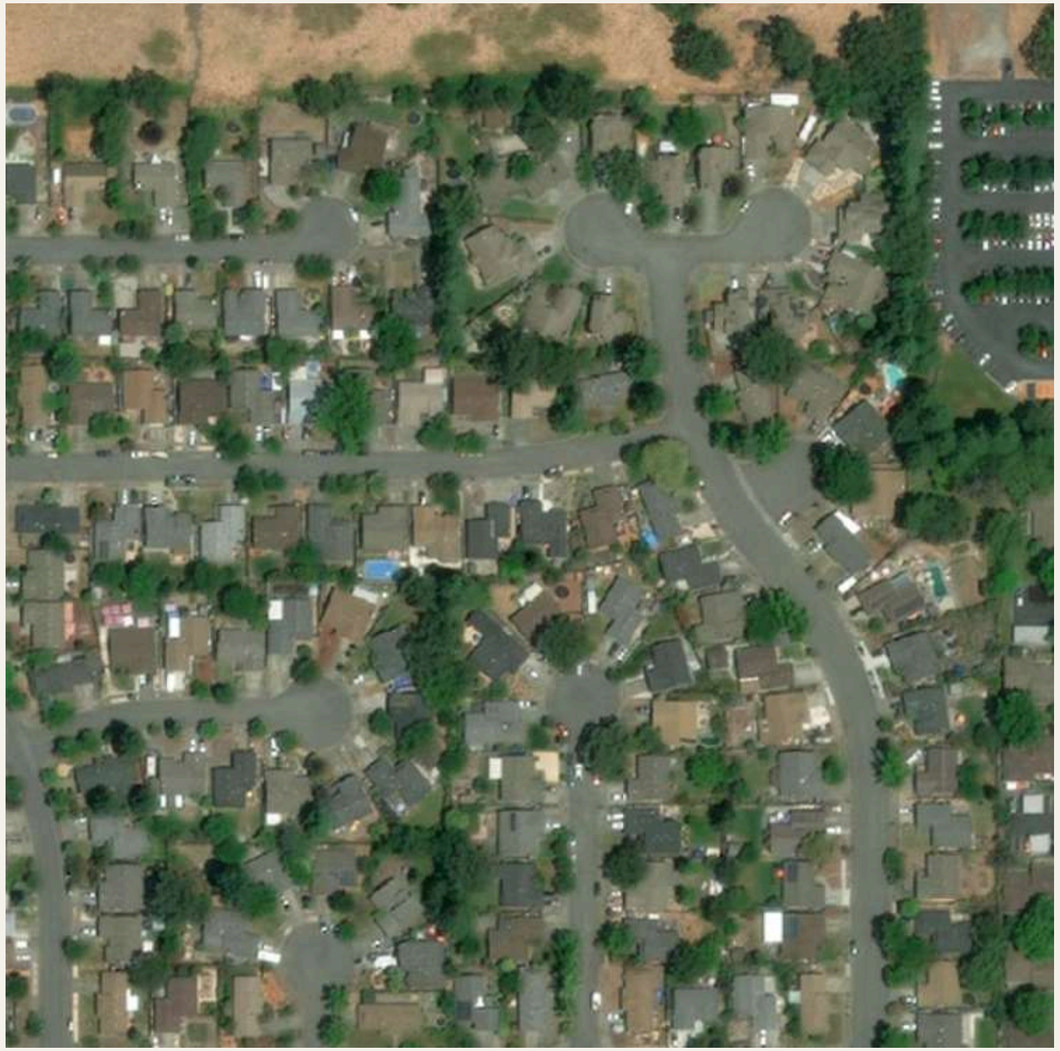
1) Satellite Image-Based Damage Assessment for Natural Disasters

- Assessing infrastructure damage after disasters using satellite images is difficult due to large areas, complex scenes, and environmental variations.
- Damage assessment is often performed manually by visual inspection of satellite images, which is slow, subjective, and not scalable.
- Rapid and automated damage assessment is required to support timely disaster response and recovery planning.
- Explore a learning-based approach to automatically assess disaster damage using pre- and post-disaster satellite image pairs.
- Learn visual change patterns related to structural damage while minimizing the influence of natural environmental changes.

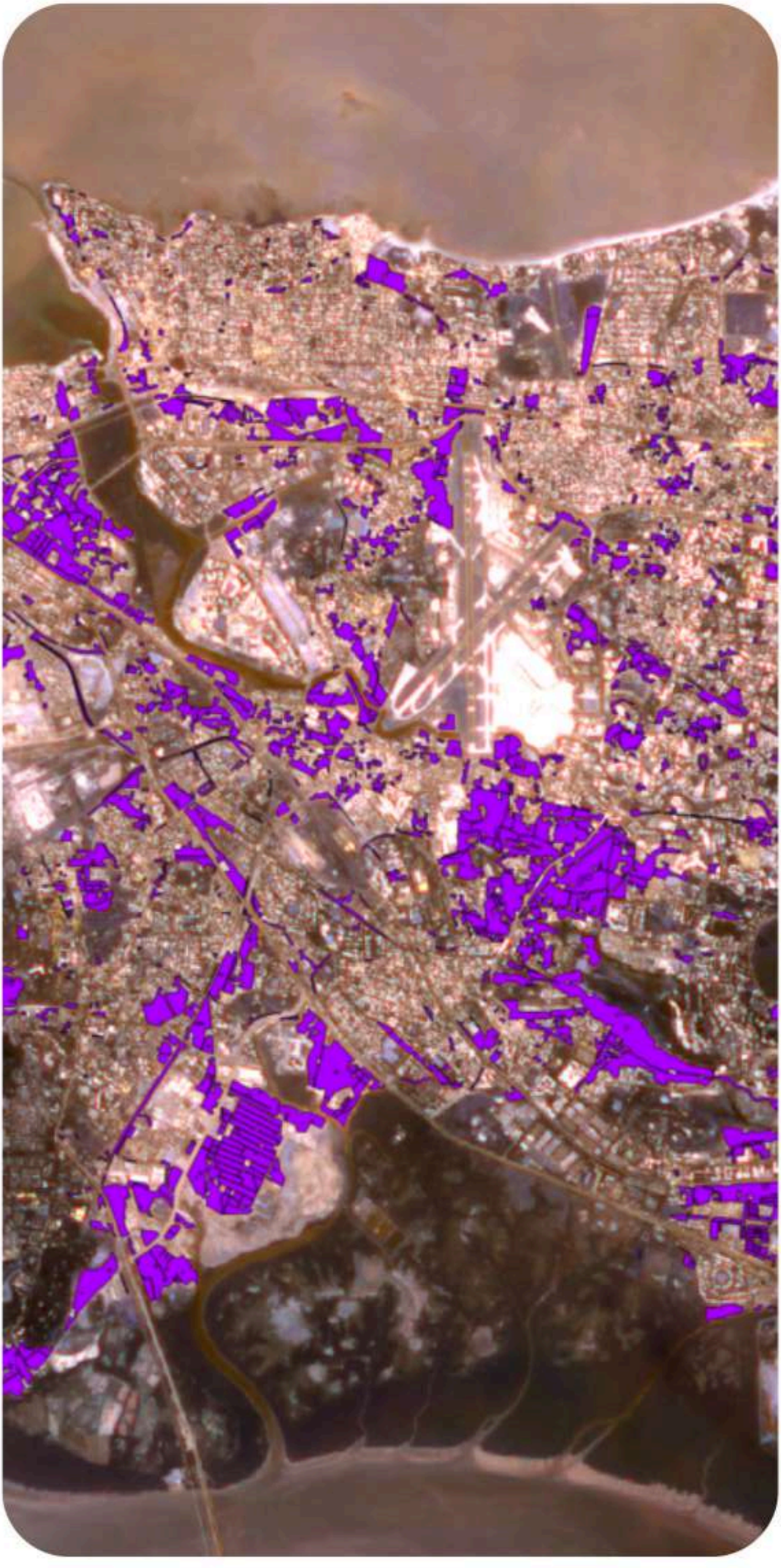
- Use the xBD (xView2) disaster damage dataset, which provides bi-temporal satellite images with damage annotations.
- Damage assessment maps highlighting affected areas or structures along with damage severity levels.
- Expected Outcome: A system demonstrating faster and more consistent damage assessment compared to manual analysis.
- Limitations: Performance may be affected by image misalignment, complex urban layouts, and visually ambiguous damage patterns.



2) Urban Vulnerability Mapping Using Satellite Images

- Urban disasters often cause disproportionate damage in poorly planned and densely populated areas, which are difficult to identify manually at scale.
- City planners and disaster authorities require early risk maps to support preparedness rather than only post-disaster analysis.
- Traditional Method: Urban vulnerability assessment is largely based on field surveys and manual mapping, which are time-consuming, subjective, and slow to update.
- Proposed Idea: Estimate urban disaster vulnerability from satellite imagery by identifying high-risk settlement patterns before disasters occur.
- Mumbai-Slum-Detection-Dataset
  https://www.kaggle.com/datasets/rupeshkumaryadav/mumbai-slum-detection-dataset

- Use satellite images to detect informal or high-density settlements at a coarse, region level rather than individual buildings. Combine visual detection results with simple spatial and mathematical measures such as density and area continuity to indicate risk.
- Risk Levels: Classify urban regions into Low, Medium, and High vulnerability zones for interpretability and practical use.
- Output: A city-wide vulnerability risk map highlighting regions with different disaster risk levels.
- Applications: Disaster preparedness, urban planning, infrastructure prioritization, and policy decision support.
- Limitations: Risk estimates depend on image resolution and may not capture socio-economic or infrastructure quality factors.



3) Probabilistic Landslide Risk Prediction Using Satellite Data and Machine Learning

- A machine learning–based system predicts the probability of landslide occurrence at a given geographic location instead of a binary yes/no output.
- Historical landslide events from the NASA Landslide Dataset are used as ground-truth labels to learn patterns associated with landslide-prone regions.
- Environmental features such as rainfall intensity and terrain slope are automatically extracted using Google Earth Engine (GEE), avoiding the need to download large satellite datasets..

- The model focuses on high-impact triggering factors (heavy rainfall and steep terrain) rather than analyzing raw satellite images, making the approach computationally efficient.
- High rainfall + steep slope → increased landslide risk
- The trained model outputs a continuous landslide risk probability, which is more informative for early warning systems and disaster preparedness.
- The system is especially useful for landslide-prone regions like Kerala, supporting disaster mitigation, planning, and response strategies.

Thank you